Beyond the latest reading.

Independent recurrent models and calibrated hybrid corrections for one-hour PM2.5 forecasts.

Measured comparisons. Validation-based selection. Explicit limitations.

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Fix the experiment before interpreting scores

The initial code and its benchmark needed separate scientific and inference corrections.

Training boundary	Fit preprocessing only on actual training windows; never learn from prediction batches.
Independent benchmark	Standalone LSTM and Bi-LSTM use the full eligible outer training window.
Honest hybrid targets	Freeze the base before a later calibration block; fit only out-of-sample residuals.
Verified inference	Share sequence-aware CLI/dashboard prediction and verify saved model hashes.

Observed history in. A future concentration out.

All hours follow the Beijing local clock; the target is one hour after the origin.

TRAIN + PREPROCESS

2010–2012 initially
Available labels only
Preserve gaps and valid peaks

SELECT ON 2013

Four expanding quarters
Screen candidates + ablations
Three predefined seeds

COMPARE ON 2014

Freeze selection first
Same observed target hours
Exploratory historical result

No target-hour measured weather. No random train/test split. No filled evaluation labels.

A sequence model can predict a level or a change

The delta formulation adds a learned correction to the latest observed concentration.

Recurrent representation

24 or 48 observed hours
LSTM or Bi-LSTM
32 or 64 hidden units per direction
Attention or final hidden states

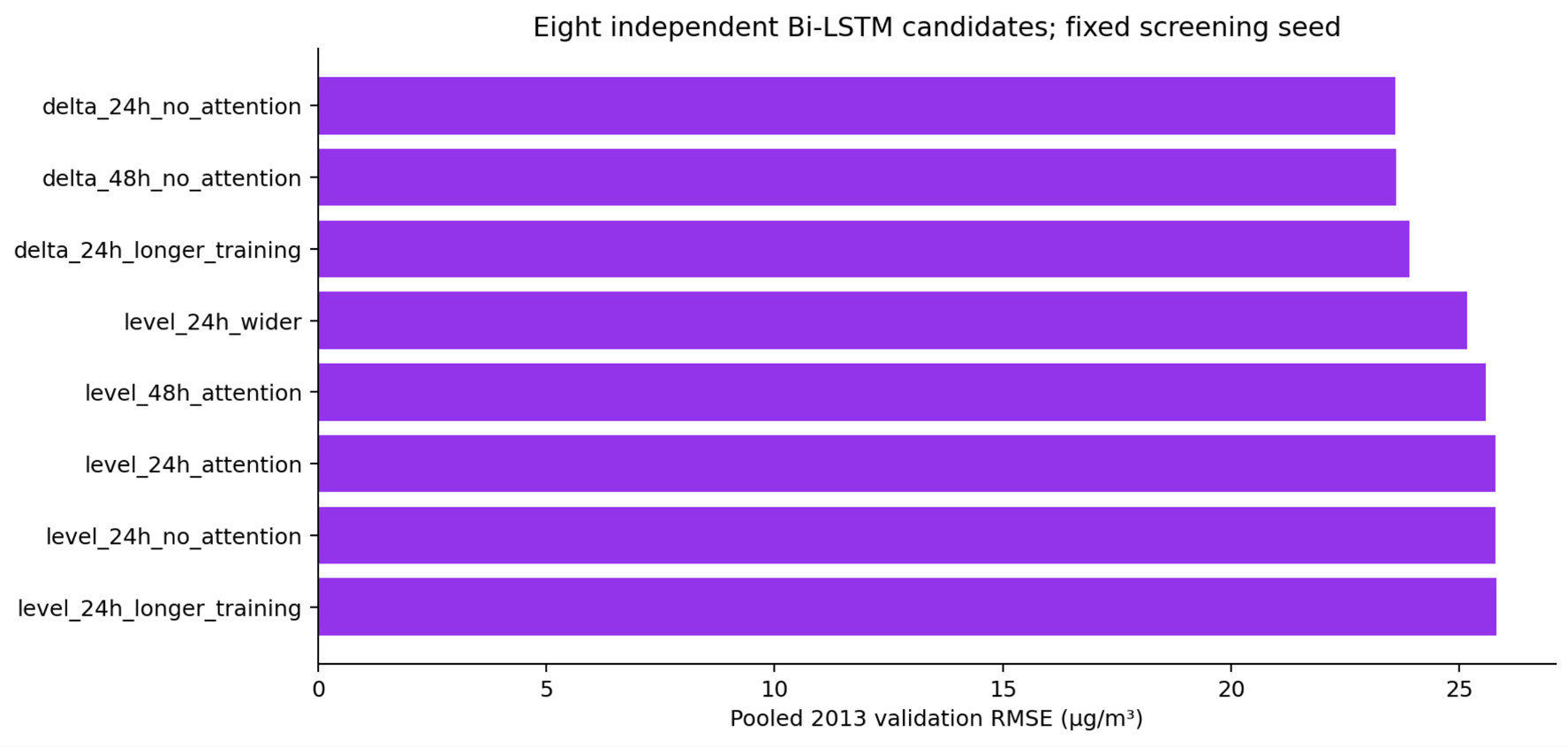
Standardized target + reconstruction

Level: next concentration
Delta: next minus current concentration
Training-only target statistics
Reconstruct and clip at zero

Without attention, both terminal Bi-LSTM states summarize the full observed window.

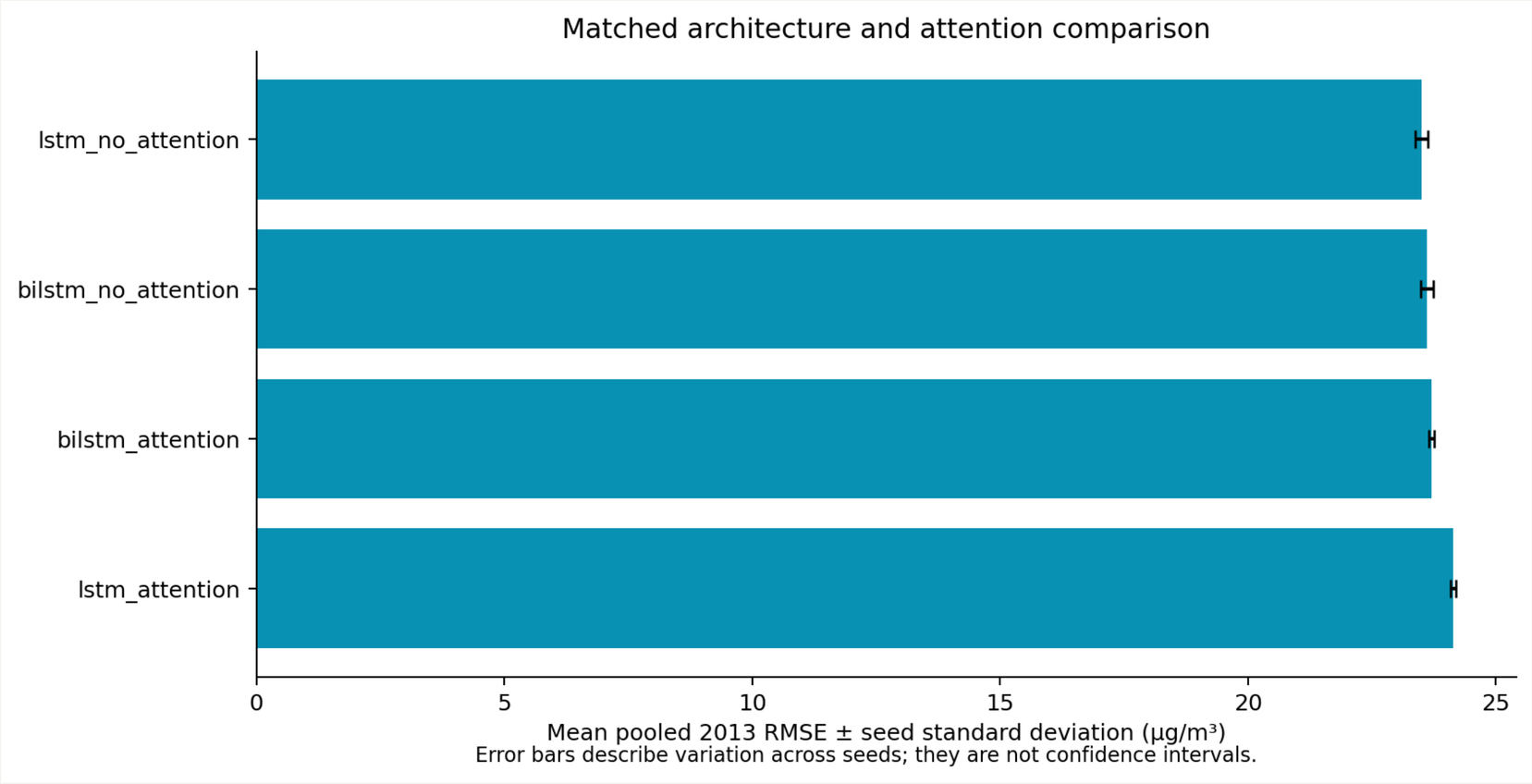
Eight candidates. One validation year.

Sequence length, width, duration, attention and target formulation; screening seed fixed at 42.



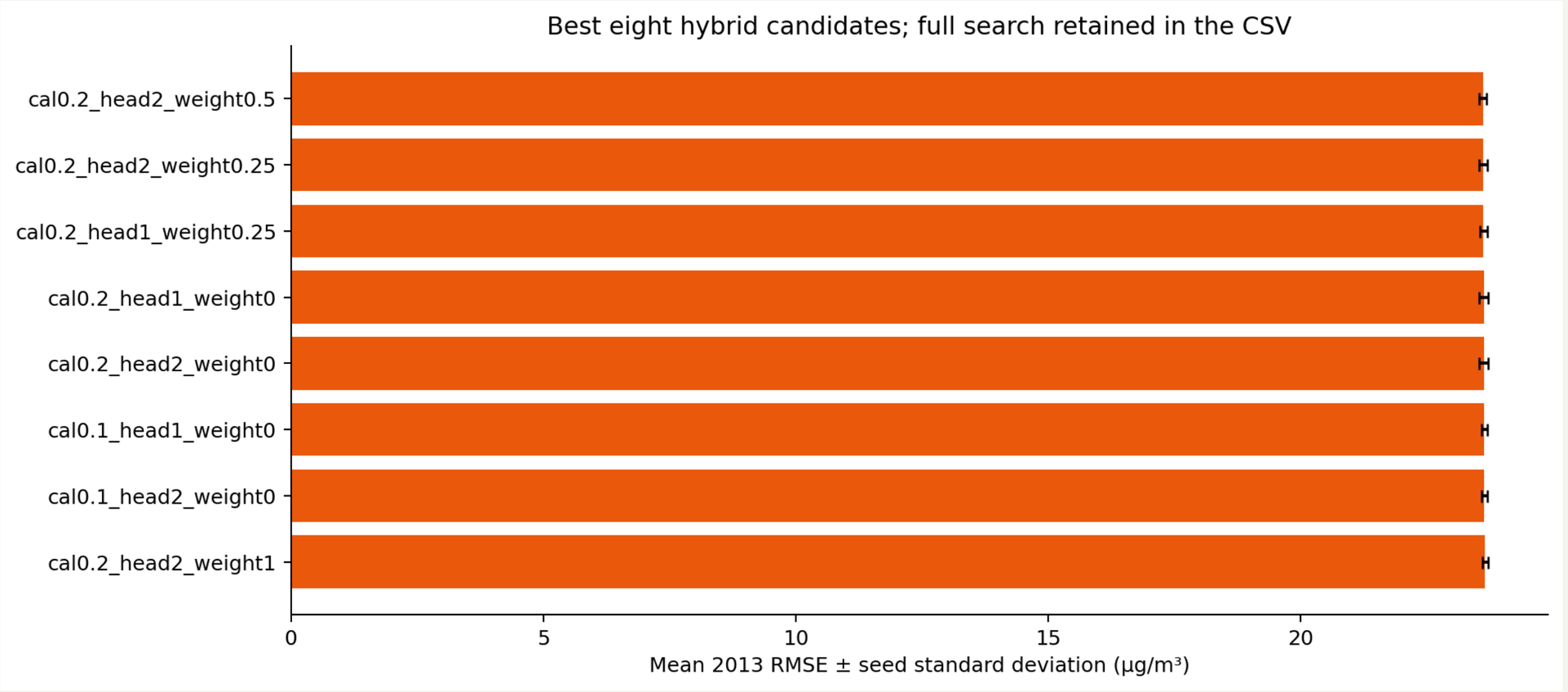
Does attention or a backward direction help?

Each matched variant is run at seeds 42, 137 and 2026 on identical validation hours.



Calibrate later. Shrink the correction.

Selected calibration: 20% of eligible origins; correction weight: 0.5.



Report every model on the same target hours

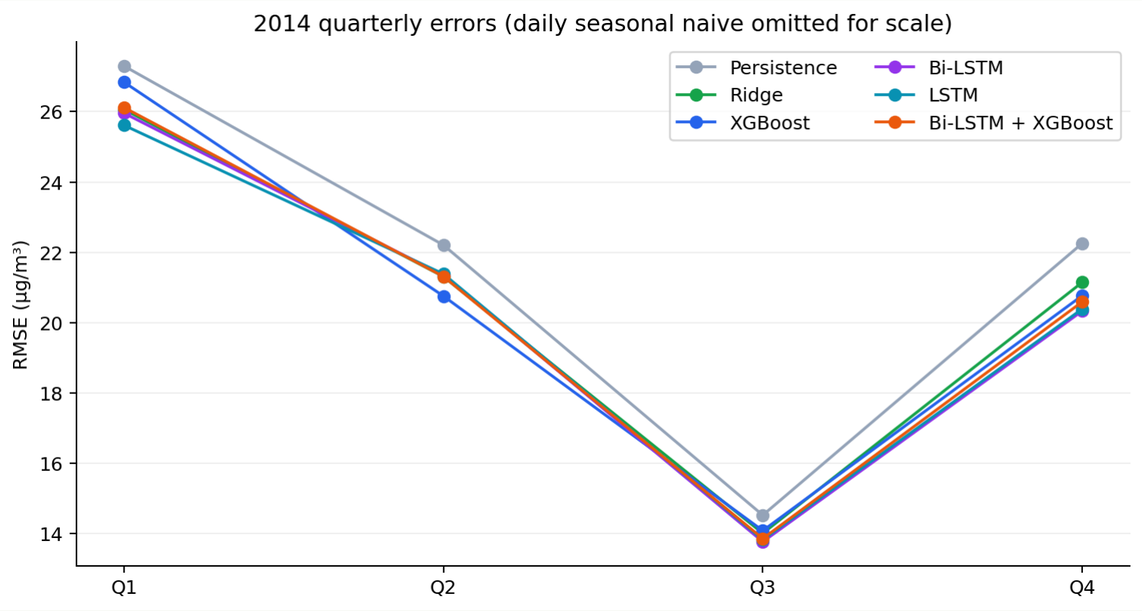
8,531 common 2014 hours. RMSE and MAE in μg/m³. Comparison seed fixed before scoring.

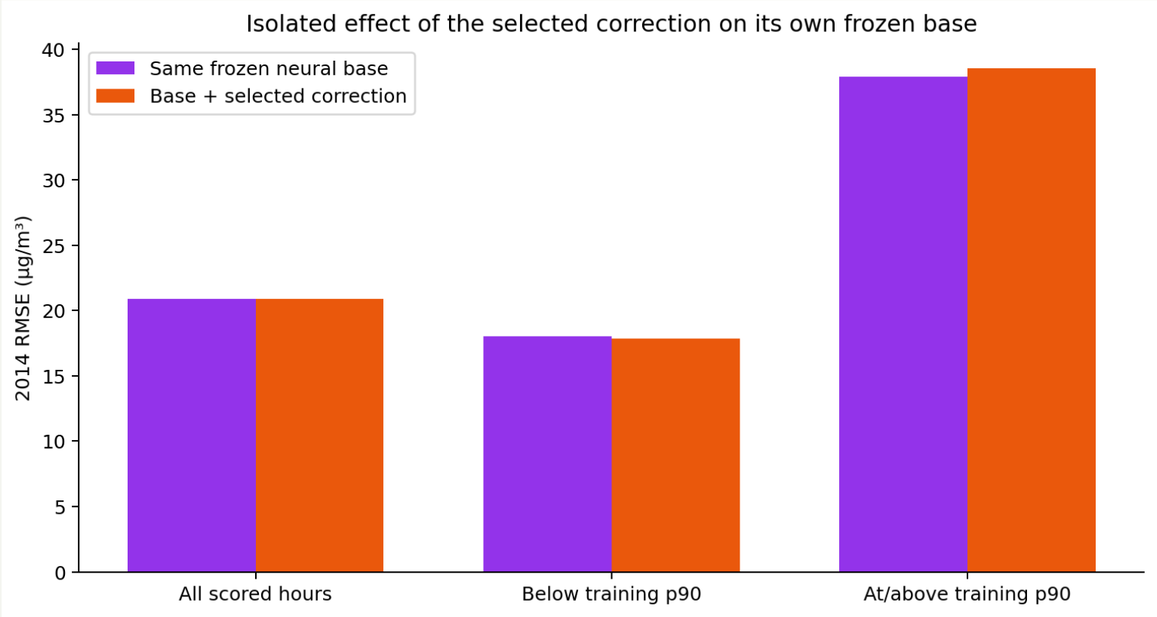
Model	MAE	RMSE	R^2
LSTM	**11.299**	**20.726**	**0.9506**
Bi-LSTM	11.457	20.799	0.9503
Bi-LSTM + XGBoost	11.483	20.919	0.9497
Ridge	11.825	21.063	0.9490
XGBoost	11.833	21.088	0.9489
Persistence	11.926	22.033	0.9442
Daily seasonal naive	67.296	98.754	-0.1205

Validation-based model preference: LSTM.

A pooled score hides changing behavior

Quarterly and pollution-regime diagnostics retain identical hours within each comparison.

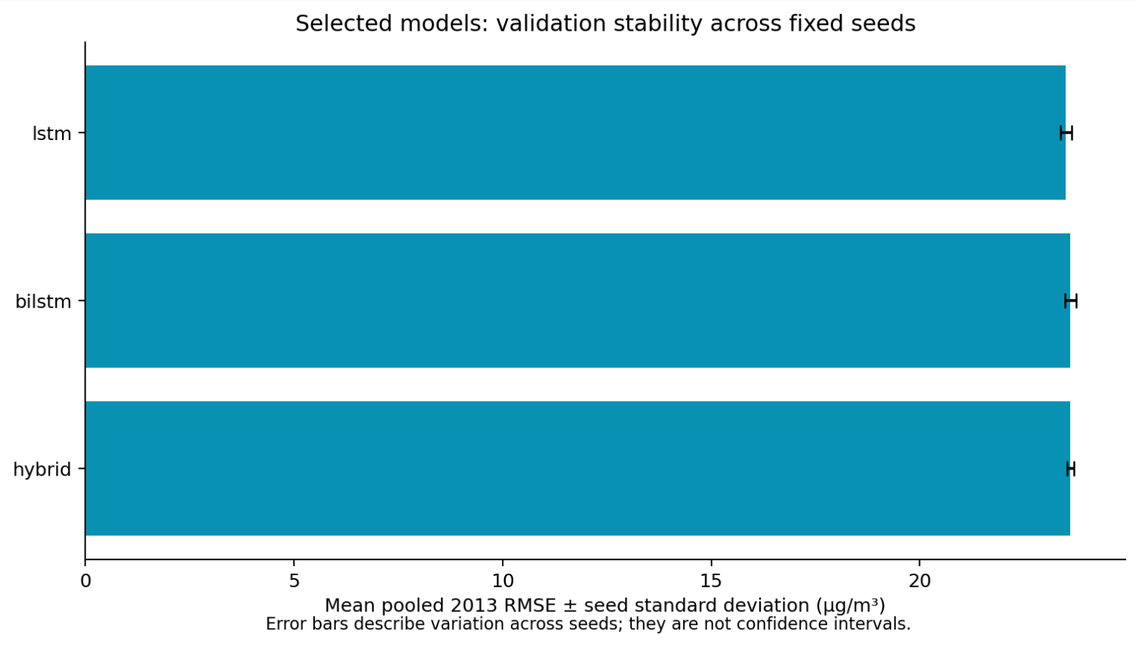


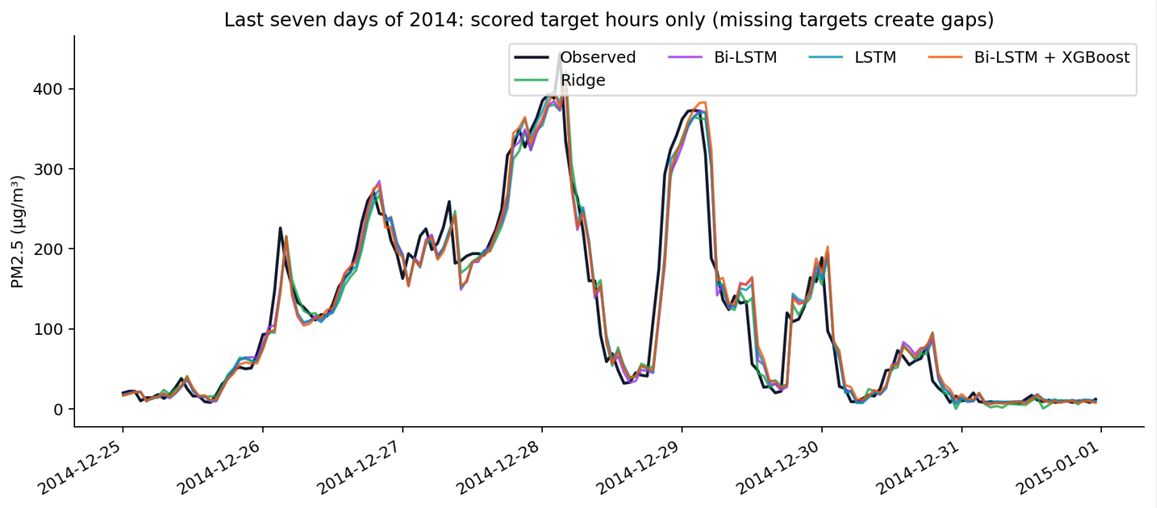


High pollution: at least 218 µg/m³, fixed from the 2010–2012 training distribution.

Repeatability and a concrete forecast sequence

The last week is selected by time; missing target hours remain visible as gaps.





Predeclared deployment seed 42. Seed variation is measured on validation.

Saved models, shared features, checked results

The notebook and package retain enough evidence to inspect and repeat the experiment.

Traceable experiment	Input/source/model hashes, frozen selection, seed scores and fold audits.
Runnable local demo	Independent Bi-LSTM, LSTM and hybrid alongside classical models.
Consistent forecasts	CLI and dashboard share features and retain 168 + 24 raw hourly slots.
Submission artifacts	IEEE report, editable slides, slide PDF, notebook and packaged weights.

The measured result is the contribution.

Validation preference: LSTM

Independent benchmarks, honest calibration, matched ablations and reproducible evidence are now in place.

2014 is exploratory. New blind accuracy claims need untouched data.

Thank you. Questions?